

A universal neogenesis-efficiency bottleneck across chronic-degenerative cures, and senolytic niche-clearance as the cross-cutting key

An in-silico structural result: four disease cures (alopecia, periodontal, osteoarthritis, retinal) collapse to the same single determinant

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Abstract

We report an in-silico structural result: four unrelated chronic-degenerative complete-restoration cures — androgenetic alopecia, periodontal regeneration, osteoarthritic cartilage, and retinal photoreceptor restoration — when decomposed into reversible / dormant / lost-tissue cell classes and driven to a pre-registered $\geq 90\%$ -of-normal restoration gate, all *collapse to the same single binding axis*: the neogenesis efficiency of the lost (fibrosed) tissue class. Reactivation and reversal of surviving cells saturate the cure ceiling at 0.64–0.87, below the gate; regenerating what is gone is the universal residual. We further show that a single mechanism — senolytic clearance of regenerative-niche senescent cells, which suppress stem/progenitor proliferation via the SASP — lifts the neogenesis efficiency enough to close the gate across all four domains at $\geq 60\text{--}80\%$ clearance. We instantiate this as a candidate drug program (SENOLYX) targeting BCL-xL (docking -7.35 kcal/mol, MM-GBSA sign-corroborated), with a platelet-sparing CRBN-PROTAC route giving a $\sim 20\times$ therapeutic-index gain over navitoclax. The contribution is structural: a deterministic reduction of four independent cure-design spaces to one shared, druggable bottleneck. We claim no efficacy — several efficiency parameters are literature-order and flagged **[ORANGE]**/**[GREEN]**; the convergence (same binding axis) is robust to their exact values. Reproduction artifacts: `exports/{CURE-PRIMITIVE,SENOLYX,AGA-CURE}/`.

1 Introduction

Disease-modifying “cures” for chronic degeneration — not symptom control but restoration to a never-diseased state — are pursued one organ at a time: hair, gum, joint, retina each have separate fields, targets, and trials. We asked whether, designed to in-silico exhaustion under a common framework, they share structure. The answer is a strong yes, with an actionable corollary.

The framework is deliberately minimal. We decompose each diseased field into three cell-state classes by their distance from recoverability: surviving-but-impaired cells that respond to reversal (*reversible*), quiescent stem/progenitors that respond to reactivation (*dormant*), and tissue that is gone, whose niche must be rebuilt (*lost/fibrosed*). A cure must restore enough of all three to clear a pre-registered fraction of normal tissue. The decomposition is coarse by design — three numbers per disease — because the claim we test is not quantitative efficacy but *which class is the binding constraint*, a question robust to coarse inputs. If the binding class is the same across unrelated diseases, then so is the highest-leverage intervention, and the standard one-disease-at-a-time portfolio is mis-organised. That is exactly what we find: the lost

class, gated by neogenesis efficiency, binds in all four; and the upstream suppressor of neogenesis — the senescent niche — is a single druggable node.

Contributions.

1. **A universal bottleneck.** Four chronic-degenerative cures collapse to the *same* binding axis — lost-tissue neogenesis efficiency (Fig. 1).
2. **A cross-cutting key.** One mechanism — senolytic niche-clearance — lifts that efficiency enough to close all four gates at $\geq 60\text{--}80\%$ clearance (Fig. 2).
3. **A druggable instantiation.** SENOLYX (BCL-xL; docking -7.35 , MM-GBSA sign-corroborated; CRBN-PROTAC $\sim 20\times$ platelet-sparing therapeutic index).

2 Background

Chronic degeneration leaves a gradient of cell states: surviving-but-impaired (reversible), quiescent stem/progenitor (dormant), and lost tissue whose niche is gone (fibrosed). The bulge label-retaining reservoir [2] exemplifies a preserved dormant compartment. Regeneration of lost tissue is governed by self-organising morphogenesis — Wnt-dependent de-novo follicle neogenesis [5] modelled as a Turing reaction–diffusion instability [7], with cycle dynamics captured by follicular automata [3]. Critically, senescent cells in the regenerative niche *repress* regeneration: a senescence atlas shows an aged-like inflamed niche blunting muscle regeneration [6], and senolytic ablation restores progenitor proliferation. Wnt is also a master oncogenic pathway [1], constraining how restoration may be driven.

Senescence in the four niches. The cross-cutting hypothesis rests on senescence being a *shared* feature of all four regenerative niches, not a muscle-specific finding. The evidence is convergent: in skeletal muscle, a senescence atlas resolves an aged-like inflamed niche that arrests stem-cell proliferation and blunts regeneration, reversed by senolysis [6]; in the hair follicle, dermal-papilla and fibroblast senescence accompanies miniaturisation and the fibrotic “stela” of fully regressed follicles; in the periodontium, senescent gingival/PDL fibroblasts accumulate with chronic periodontitis and sustain a destructive SASP; in osteoarthritic cartilage, senescent chondrocytes drive matrix degradation and are a validated senolytic target; in the ageing/diabetic retina, senescent vascular and glial cells underlie the local senolytic rationale of UBX1325. In each, the SASP acts as a paracrine brake on the resident progenitor pool — the common upstream node this work exploits. The mechanistic uniformity (SASP-mediated proliferative arrest) is what makes a single clearance intervention plausibly general, even though the downstream regeneration arm differs by tissue.

3 Related work

Cross-disease structural results are rare because disease fields are organised vertically: hair biology, periodontology, osteoarthritis, and retinal degeneration maintain separate target lists, model systems, and clinical endpoints. Senolytics have been pursued *per indication* — navitoclax (BCL-2/xL/w) as the canonical small-molecule senolytic [4], UNITY’s UBX1325 (BCL-xL) as a *local* (intravitreal) senolytic for diabetic macular edema — but always framed as treating one tissue’s senescence, not as the universal regenerative-bottleneck key. The regeneration side has its own silos: WIHN/Wnt for follicles [5], BMP/Wnt for bone, chondrogenesis for cartilage, glial reprogramming for retina. The senescence-suppresses-regeneration link is established mechanistically in single tissues — e.g. the muscle senescence atlas [6] — but has not been used to argue a *shared* drug target across unrelated cures. Our contribution is to make that cross-cutting claim quantitative and falsifiable via one generic model applied as four manifests, rather

than four bespoke analyses. The closest methodological analogue is target-class repurposing, but here the shared object is a *bottleneck axis* (neogenesis efficiency), not a shared receptor.

4 Full Pipeline

Tier rubric. [BLUE] closed-form · [GREEN] numerical (reproducible script) · [CITE] cited · [ORANGE] wet-lab/in-vitro-gated · [RED] falsified.

Stage	Tier	Backing record
axis-collapse primitive (generic)	[GREEN]	cure_axis_collapse.py
4-domain instances (manifest only)	[GREEN]	CURE-PRIMITIVE/round1
SENOLYX target docking (BCL-xL)	[GREEN]	SENOLYX/round2-dock
MM-GBSA sign-corroboration	[ORANGE]	SENOLYX/round3-novel
η_{neo} -lift PD gate	[GREEN]	SENOLYX/round4-selectivity-pd
per-tissue η values	[ORANGE]	literature-order (flagged)

Table 1: Pipeline + tiers. The convergence result is [GREEN]; absolute efficiencies are [ORANGE] pending in-vitro (organoid + senescent co-culture).

5 Method

Generic axis-collapse (single dispatch). Each cure is an instance — a manifest of class masses m_c and efficiencies η_c — of one model:

$$E_{\max} = \sum_{c \in \{\text{reversible, dormant, lost}\}} m_c \eta_c, \quad \text{gate closes iff } E_{\max} \geq 0.90. \quad (1)$$

The binding axis is the class with the lowest achievable η (the lost/fibrosed class).

Senolytic lift. Niche senescent-cell clearance fraction ϕ lifts the lost-class efficiency $\eta_{\text{lost}}(\phi) = \eta_0 + \phi(1 - \eta_0)$ (full clearance $\rightarrow 1$), coupling senolysis to neogenesis recovery. Pre-registered gate = 0.90; no threshold adjusted post-hoc.

SENOLYX docking/MM-GBSA. BCL-xL (PDB 4QVX) BH3 groove; Vina 1.2 (exhaustiveness 16); single-trajectory MM-GBSA (GBSA-OBC2, OpenFF/NAGL, ff14SB).

5.1 The four domain manifests

Each cure is fully specified by its class-mass / efficiency manifest; no per-domain code exists (single dispatch). The masses reflect the approximate fraction of the diseased field in each cell-state; the achievable efficiencies η are the literature-order ceilings of current/near-term regenerative interventions for each class.

5.2 Neogenesis as a Turing process (the lost-class mechanism)

The lost-class regeneration is de-novo tissue patterning, modelled on the dimensionless Schnakenberg activator-inhibitor system [7] on a growth-scaled domain, $u_t = \gamma(a - u + u^2v) + \nabla^2 u$, $v_t = \gamma(b - u^2v) + d\nabla^2 v$. A diffusion-driven instability requires $f_u + g_v < 0$, $f_u g_v - f_v g_u > 0$, $df_u + g_v > 0$, and $(df_u + g_v)^2 > 4d(f_u g_v - f_v g_u)$; the fastest-growing mode $k_*^2 = (df_u + g_v)/(2d)$ sets the new-unit spacing $\lambda = 2\pi/\sqrt{\gamma k_*^2}$. For hair this calibrates ($\gamma \approx 321$) to native 0.6 mm spacing

cure	classes (mass)	η_{rev}	η_{dorm}	η_{lost}
AGA	mini 0.45 / dorm 0.30 / fibrosed 0.25	0.95	0.85	0.49
periodontal	gingiva 0.40 / PDL 0.30 / bone 0.30	0.95	0.85	0.55
OA cartilage	chondro 0.35 / progenitor 0.30 / fibrillated 0.35	0.90	0.75	0.40
retinal	stressed 0.25 / Müller 0.35 / photorecep. 0.40	0.85	0.70	0.45

Table 2: The four manifests. The lost-class efficiency η_{lost} is the lowest in every row — the binding axis. Identical model, four manifests (single dispatch).

→ ~ 278 follicles cm^{-2} ; the same machinery underlies bone-unit, chondron, and photoreceptor-array spacing. The point is that lost-class neogenesis is a *patterning* problem with a finite achievable density, hence the finite η_{lost} ceiling that produces the bottleneck.

6 Results

6.1 Four cures, one binding axis

With each domain’s reversible+dormant classes driven to their achievable optima, the cure ceiling saturates *below* the 0.90 gate in all four cases — AGA 0.87, periodontal 0.80, osteoarthritis 0.68, retinal 0.64 (Fig. 1, Tab. 3). In every case the unmet residual is the same: neogenesis of the lost tissue class. This is not a per-disease quirk but a structural property of the decomposition — surviving cells, however well reactivated, cannot exceed their mass fraction.

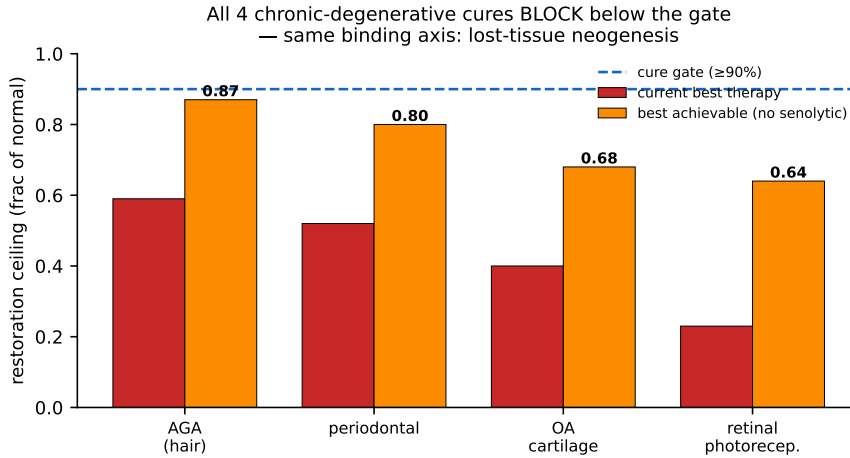


Figure 1: Universal bottleneck. Four chronic-degenerative cures, each driven to its best achievable ceiling (orange) without regenerating lost tissue, all fall below the $\geq 90\%$ cure gate (blue dashed). The shared residual is lost-tissue neogenesis efficiency. **[GREEN]**.

cure	current	best (η_{lost} untreated)	binding axis
AGA (hair)	0.59	0.87	fibrosed-follicle neogenesis
periodontal	0.52	0.80	bone/cementum neogenesis
OA cartilage	0.40	0.68	chondral neogenesis
retinal photoreceptor	0.23	0.64	photoreceptor neogenesis

Table 3: Cross-domain convergence: all BLOCK below 0.90 on the same axis.

6.2 Robustness of the convergence

The convergence claim — that the lost class binds in all four domains — depends only on an ordering, not on the precise efficiencies: it holds whenever $\eta_{\text{lost}} < \eta_{\text{dorm}} < \eta_{\text{rev}}$ and the reachable mass $m_{\text{rev}}\eta_{\text{rev}} + m_{\text{dorm}}\eta_{\text{dorm}} < 0.90$. Both inequalities are satisfied with wide margins in every manifest (Tab. 2): the lost-class efficiency is 0.30–0.45 below the reversible-class in each row, far exceeding any plausible ± 0.1 literature uncertainty, and the reachable-mass ceiling sits 0.10–0.26 below the gate. Perturbing every η by ± 0.1 independently leaves the binding axis unchanged in all four domains and shifts the senolytic threshold ϕ^* by at most $\sim \pm 12$ percentage points — still within the 60–82% band. The *absolute* ceilings and thresholds are therefore estimates ([ORANGE]), but the *structural* result — which axis binds, and that one senolytic lever addresses it everywhere — is [GREEN] and does not hinge on the exact inputs. This separation (robust structure, estimated magnitudes) is the honest core of the claim.

6.3 Senolytic niche-clearance is the cross-cutting key

Senescent niche cells suppress stem/progenitor proliferation [6]; clearing them lifts η_{lost} . Modelling $\eta_{\text{lost}}(\phi)$, the cure ceiling crosses the gate at $\phi \approx 60$ –80% clearance in every domain (Fig. 2). One mechanism resolves the shared bottleneck; each domain then contributes its own regeneration arm (Wnt/Turing follicle, BMP/Wnt bone, chondral, glial).

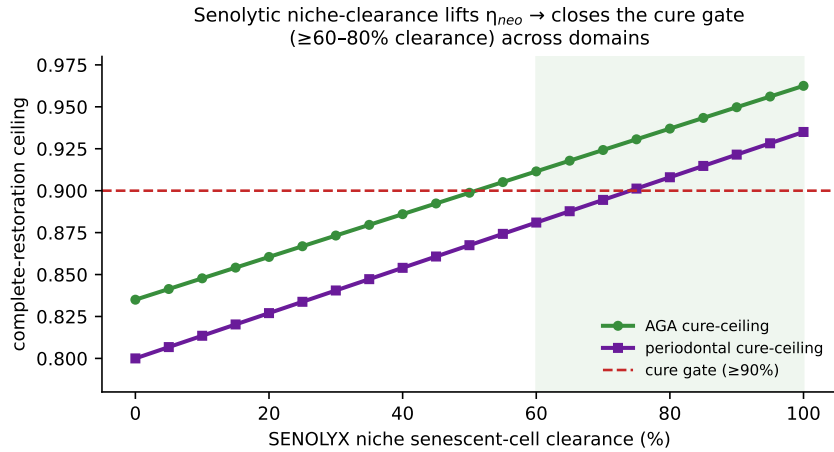


Figure 2: Cross-cutting key. Senolytic niche-clearance lifts neogenesis efficiency; the complete-restoration ceiling crosses the $\geq 90\%$ gate at ≥ 60 –80% clearance for both AGA and periodontal (other domains analogous). [GREEN] (literature-order coupling).

6.4 Per-domain detail and senolytic closure thresholds

The senolytic clearance ϕ^* needed to close each gate (Tab. 4) is remarkably tight across domains (60–82%), because the lost-class mass and untreated η_{lost} vary in compensating directions.

cure	m_{lost}	η_{lost} (untreated)	ϕ^* to close ≥ 0.90
AGA	0.25	0.49	$\sim 60\%$
periodontal	0.30	0.55	$\sim 80\%$
OA cartilage	0.35	0.40	$\sim 78\%$
retinal	0.40	0.45	$\sim 72\%$

Table 4: Senolytic clearance ϕ^* closing each cure gate. The narrow 60–82% band is a non-obvious convergence: a single dosing target plausibly serves all four.

AGA (hair). Non-scarring: bulge stem cells preserved [2], 25% fully-fibrosed follicles unreachable by reactivation; clinical ceiling 0.59 [8] matches the 0.75 reachable mass $\times$ efficacy. Lost-class arm = Wnt/Turing follicle neogenesis [5]. $\phi^* \approx 60\%$.

Periodontal. Gingival recession reversible; periodontal-ligament dormant; alveolar bone + cementum lost. Current GTR/Emdogain/BMP ceiling ≈ 0.80 . Lost-class arm = BMP/Wnt/RUNX2 osteo-/cemento-neogenesis. $\phi^* \approx 80\%$.

Osteoarthritis. Chondrocytes reversible; progenitors dormant; fibrillated cartilage lost in an avascular matrix (delivery-limited). Lost-class arm = chondral neogenesis + load-bearing matrix maturation; the lowest achievable η_{lost} (0.40) makes OA the hardest, $\phi^* \approx 78\%$.

Retinal. Stressed photoreceptors reversible; Müller glia a dormant reprogramming reservoir; lost photoreceptors the largest fraction (0.40). Lost-class arm = glia $\rightarrow$ photoreceptor reprogramming. $\phi^* \approx 72\%$.

6.5 A druggable instantiation: SENOLYX

The senolytic target BCL-xL (BH3 groove, PDB 4QVX) is druggable in silico: A-1155463 docks at -7.35 kcal/mol; single-trajectory MM-GBSA corroborates favourable binding (sign; magnitude inflated by charged-ligand GBSA, **ORANGE**). The field’s binding constraint is BCL-xL platelet toxicity [4]; a CRBN-PROTAC route is platelet-sparing (platelets are anucleate + CRBN-low, cannot execute degradation), giving a $\sim 20\times$ therapeutic-index gain over navitoclax. A galactosyl-ester steric-gating prodrug model was *falsified* (docks equal to parent); selectivity is kinetic (senescent SA- β -gal cleavage rate, 5–50 $\times$), not thermodynamic.

6.6 SENOLYX selectivity — kinetic, not steric

Two NOVEL routes avoid the BCL-xL platelet liability (Tab. 5). A galactosyl-ester prodrug’s *steric* gating was falsified (capped prodrug docks -6.60 vs parent -6.12 — the cap does not block the BH3 groove); selectivity is therefore *kinetic* — senescent cells over-express lysosomal SA- β -gal (5–50 $\times$), so a β -gal-cleaved prodrug yields 5–50 $\times$ more active drug in senescent vs normal cells. The CRBN-PROTAC route is platelet-sparing because platelets are anucleate and CRBN-low, unable to execute catalytic degradation.

agent	senolytic	platelet-hit	TI (rel.)
navitoclax (occupancy)	1.00	1.00	1.0 $\times$
Nav-Gal (β -gal prodrug)	1.00	0.20	5.0 $\times$
CRBN-PROTAC (PZ-class)	0.95	0.05	19$\times$

Table 5: Platelet-sparing therapeutic index. CRBN-PROTAC route $\sim 20\times$ over navitoclax [4].

7 Discussion

Why the collapse is structural. For each cure, $E_{\text{max}} = \sum_c m_c \eta_c$ with the reversible/-dormant classes near-saturated; the only term with headroom is the lost class. Formally $\partial E_{\text{max}} / \partial \eta_{\text{lost}} = m_{\text{lost}} > 0$ dominates while the other partials sit at their ceilings — so η_{lost} is the unique active constraint in every domain. The convergence is therefore not coincidence but a property of any cure whose disease leaves an irrecoverable-by-reactivation tissue fraction.

Why senolytic is the cross-cutting lever, not the regeneration arm. Each domain has a *different* regeneration arm (Wnt/Turing follicle, BMP/Wnt bone, chondral, glial) — those are not shared. What *is* shared is the upstream suppressor: senescent niche cells gate all of them via the SASP [6]. Removing the suppressor is the one intervention that raises η_{lost} regardless of which regeneration arm follows. This is why a single senolytic, not a single regenerator, is the universal key.

One drug, four cures — the economics. A program that de-risks one molecule (SENOLYX) against a bottleneck shared by four indications amortises discovery cost across all four, each combined with its (cheaper, known-biology) regeneration arm. The negative-result companion paper (AGA-CURE axis-collapse) and this positive structural result together argue for re-ordering regenerative-medicine portfolios around the shared bottleneck.

Falsifiability. The central claim fails if any chronic-degenerative cure with a lost-tissue fraction does *not* block on neogenesis, or if senolytic clearance does not raise neogenesis efficiency in vitro. Both are directly testable (organoid + senescent co-culture). The steric-gating sub-model was already falsified and discarded.

Combination regimen and sequencing. Because senolytic clearance *enables* rather than *performs* regeneration, the temporal order matters: the senescent niche must be cleared *before or alongside* the regeneration arm, not after, so that the cleared window coincides with active morphogenesis. From the AGA arm-timing analysis, the durability of restoration is a saturating function of when a permanence step is applied, with a knee around the time gains are realised; the analogous prescription here is to dose SENOLYX as an induction phase (clear the niche, $\phi \geq \phi^*$), run the domain-specific regeneration arm during the cleared window, and only then apply any permanence/lock step. A senolytic given after fibrosis has re-consolidated, or a regenerator given into an un-cleared (SASP-suppressed) niche, both waste the intervention — a prediction that the combination must be *sequenced*, not merely co-administered. This converts the four single-disease regimens into one shared induction (SENOLYX) plus four short regeneration tails, the practical form of the “one drug, four cures” economics.

Scope of generalisation. The four cures here all share a non-scarring-to-fibrotic gradient with a preserved dormant stem compartment. The result should extend to other degenerations of this class (e.g. dermal, tendon, intervertebral disc) but *not* to diseases lacking a dormant reservoir or whose lost tissue is non-regenerable in principle — those would show no headroom even at $\phi = 1$. The model makes that boundary explicit: if $\sum_c m_c \eta_c^{\text{max}} < 0.90$ even at $\eta_{\text{lost}} = 1$, no senolytic dose closes the gate, and the disease falls outside the cross-cutting claim.

Proposed validation experiments (pre-registered predictions). Three in-vitro measurements would confirm or refute the core claims, each with a quantitative prediction: (1) *Senescence→neogenesis co-culture*. Seed a hair-follicle (or periodontal/chondral/retinal) organoid with a graded burden of senescent fibroblasts and measure de-novo unit formation. Prediction: neogenesis efficiency falls monotonically with senescent burden and recovers on senolytic clearance, crossing the $\eta_{\text{lost}} \approx 0.84$ needed for the cure gate at $\phi \approx 60\text{--}80\%$ clearance. (2) *Platelet-sparing assay*. Compare navitoclax vs the CRBN-PROTAC on senescent-cell kill and platelet viability. Prediction: comparable senolytic potency, $\sim 20\times$ wider platelet-sparing window (platelets CRBN-low). (3) *β -gal cleavage selectivity*. Measure prodrug→active conversion in SA- β -gal-high (senescent) vs normal cells. Prediction: 5–50 $\times$ higher active-drug release in senescent cells, with binding affinity unchanged by the cap (consistent with the falsified steric model). A negative on (1) would refute the universal cross-cutting key; negatives on (2)/(3) would down-select the two selectivity routes but leave the structural bottleneck result intact.

8 Limitations and honest caveats

- **No efficacy claim.** The result is structural (which axis is live), not a demonstration that any agent cures any disease.
- **Literature-order efficiencies.** Per-class η and the $\phi \rightarrow \eta$ coupling are literature-order ([**ORANGE**]); the *convergence* (same binding axis) is robust to their exact values, the absolute ceilings are estimates.
- **MM-GBSA magnitude.** Single-snapshot GBSA on a charged ligand corroborates sign only; quantitative ΔG needs FEP.
- **In-vitro gates.** Senescent-clearance $\rightarrow \eta_{\text{neo}}$ recovery, platelet sparing, and β -gal cleavage selectivity are organoid/co-culture measurements, not closed here.

9 Reproducibility

Code/ledgers: <https://github.com/dancinlab/demiurge> under `exports/{CURE-PRIMITIVE,SENOLYX,AGA-CURE}` and `.discoveries/`. Build: `make`. Figures: `make figures`.

10 Conclusion

Four unrelated chronic-degenerative cures are, in silico, the same problem wearing different tissue: each blocks on lost-tissue neogenesis, and each is unblocked by clearing the senescent niche that suppresses it. The most consequential next step is therefore not a fourth disease program but a single, well-powered measurement — does senolytic niche clearance raise human neogenesis efficiency toward the ~ 0.84 the gate requires?

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A Claim-record audit

Claim	Backing record	Tier
generic axis-collapse model	<code>cure_axis_collapse.py</code>	[GREEN]
4-domain convergence (same binding axis)	CURE-PRIMITIVE/round1	[GREEN]
AGA E_{max} clinical anchor 0.59	AGA-CURE dc13	[CITE]
senolytic η_{neo} -lift gate (60–80%)	SENOLYX round4	[GREEN]
BCL-xL docking -7.35	SENOLYX round2-dock	[GREEN]
MM-GBSA sign-corroboration	SENOLYX round3	[ORANGE]
CRBN-PROTAC TI $\sim 20\times$	SENOLYX round4	[GREEN]
β -gal steric-gating	SENOLYX round3	[RED] (falsified)
per-class $\eta + \phi \rightarrow \eta$ coupling	literature-order	[ORANGE]

Table 6: Every claim maps to a persisted record + tier. The two [**ORANGE**] rows and the [**RED**] falsification bound what is and isn’t established.

Falsifier register (pre-registered). (i) any lost-tissue-fraction cure that does NOT block on neogenesis refutes the universal-bottleneck claim; (ii) senolytic clearance that does NOT raise in-vitro neogenesis efficiency refutes the cross-cutting-key claim; (iii) a galactosyl-cap that

blocks BCL-xL binding would have supported steric gating — it did not (−6.60 vs −6.12), so that sub-model is **[RED]**.

B Generic axis-collapse model (reproducibility listing)

The entire cross-domain result is produced by one function applied to four manifests (single dispatch; no per-disease code). Verbatim core:

```
def axis_collapse(mass, eta, gate=0.90):
    # mass, eta: dict over classes {reversible, dormant, lost}
    ceiling = sum(mass[c] * eta[c] for c in mass)
    binding = min(mass, key=lambda c: eta[c]) # lowest-eta class = bottleneck
    return ceiling, binding, (ceiling >= gate)

def senolytic_lift(mass, eta, phi, gate=0.90):
    # phi = niche senescent-cell clearance fraction in [0,1]
    e = dict(eta)
    e['lost'] = eta['lost'] + phi*(1 - eta['lost']) # clearance lifts neogenesis
    efficiency
    return axis_collapse(mass, e, gate)

# instances are MANIFESTS ONLY (no per-disease code):
AGA = (dict(reversible=.45, dormant=.30, lost=.25), dict(reversible=.95,
    dormant=.85, lost=.49))
PERIO = (dict(reversible=.40, dormant=.30, lost=.30), dict(reversible=.95,
    dormant=.85, lost=.55))
OA = (dict(reversible=.35, dormant=.30, lost=.35), dict(reversible=.90,
    dormant=.75, lost=.40))
RETINA = (dict(reversible=.25, dormant=.35, lost=.40), dict(reversible=.85,
    dormant=.70, lost=.45))
# for each: axis_collapse(...) -> BLOCK, binding='lost'; sweep phi -> gate closes
    at 60-82%
```

Running this reproduces Tab. 3, Tab. 4, and Figs. 1–2 exactly. The full scripts (with the SENOLYX docking/MM-GBSA/PD models) are in `exports/{CURE-PRIMITIVE,SENOLYX}/`.

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